

August 21, 2026

Dear BigGeo Hiring Team,

I am writing to apply for the Spatial Data Onboarding Specialist position at BigGeo. My M.Sc. research at the University of Calgary was already closely connected to BigGeo through a BigGeo/Mitacs research software role, where I worked on scalable management of Earth observation data. Because of that experience, this role feels like a very natural next step: preparing, validating, documenting, and integrating spatial datasets so they can become reliable infrastructure for real-world spatial intelligence.

In my research software work, I built C++ and Python systems for processing, encoding, storing, and retrieving satellite-imagery data. I automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs, then developed DGGS-based workflows that converted imagery into structured tensor files for multiresolution access. That work required careful attention to data quality, spatial structure, processing performance, metadata context, and whether downstream users could retrieve the right data reliably.

I also developed a digital elevation model super-resolution framework using Python, PyTorch, Rasterio, Google Earth Engine, and Streamlit. I collected and preprocessed terrain datasets, evaluated outputs using terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors, and built visualization tools to inspect model behavior. This strengthened my ability to notice when spatial data or model outputs are not behaving as expected, which is central to the quality-gate responsibility described in the posting.

Beyond geospatial research, I bring backend and data workflow experience with Django REST APIs, PostgreSQL, Git, Linux, Docker, AWS-based coursework, and GitHub Actions. I have also been building agentic AI workflows for job matching and application preparation, using LLMs to assist with structured review, decision support, and documentation while keeping human validation in the loop. That maps well to BigGeo's emphasis on using AI to accelerate inspection, schema discovery, metadata drafting, and repeatable onboarding steps without sacrificing accuracy.

What excites me most is the practical importance of the role. Spatial data is only useful when it is clean, structured, documented, indexed, and trusted. I would be grateful for the opportunity to bring my BigGeo-adjacent research background, spatial data engineering experience, and careful documentation habits into a role that directly shapes the quality and reliability of the Spatial Cloud data ecosystem.

Thank you for considering my application. I look forward to the opportunity to discuss how my geospatial software, remote-sensing, validation, and AI-assisted workflow experience can contribute to BigGeo.

Sincerely,
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